

Smart Home Media Server

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Project Summary

- Built a self-hosted home server on Ubuntu Server
- Centralized media streaming, cloud storage, and monitoring
- Used Docker to deploy and manage services
- Added Raspberry Pi integration for physical status indicators
- Designed the system to be expandable for automation and future smart-home features

Problem

Many families rely on multiple separate platforms for:

- media streaming
- file storage
- remote access

Problems:

- Repeat payments
- reduced privacy and control of data
- limited customization

Solution

My solution was to build a Docker-based smart home media platform with:

- Plex for local and remote media access
- Nextcloud for personal cloud storage
- Nginx Proxy Manager for reverse-proxy/domain access for both public-facing services
- Cloudflare Tunnel for domain-based external routing
- Prometheus + Grafana for monitoring
- node_exporter + cAdvisor for host and container metrics
- Raspberry Pi 4 for system status output

Architecture

Internet

- └─ cloud.arnzenserver.org → Cloudflare Tunnel → Nginx Proxy Manager → Nextcloud
- └─ media.arnzenserver.org → Cloudflare Tunnel → Nginx Proxy Manager → Plex

Ubuntu Server

- └─ Plex
- └─ Nextcloud
- └─ Nginx Proxy Manager
- └─ Cloudflared Tunnel
- └─ Prometheus
- └─ Grafana
- └─ node_exporter
- └─ cAdvisor

Raspberry Pi 4

- └─ Python LED status script
- └─ Physical server-health indicators



Technology Stack

- Ubuntu Server
- Docker / Docker Compose
- Plex
- Nextcloud
- Nginx Proxy Manager
- Prometheus
- Grafana
- node_exporter
- cAdvisor
- Raspberry Pi 4
- Python 3
- gpiozero

What Works

- Plex local access works
- Plex external access works through `media.arnzenserver.org`
- Nextcloud access works through `cloud.arnzenserver.org`
- Cloudflare Tunnel routing is working
- Nginx Proxy Manager routing is working
- Prometheus is collecting metrics
- Grafana dashboards are working
- Host and container monitoring are verified
- Raspberry Pi LED hardware wiring is working and ready for demo

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Ctrl K

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 Overview

 Recents >

 AI Crawl Control >

 Log Explorer >

 Analytics & logs >

 DNS >

Records

Analytics

Settings

 Email >

 SSL/TLS >

 Security >

 Access >

 Speed >

 Caching >



arnzenserver.org [☆](#) [Free](#) [✦ Ask AI](#) [🗣 Support](#) [👤](#)

DNS management for
arnzenserver.org

Review, add, and edit DNS records. Changes propagate globally within minutes.

DNS Setup: Full ⓘ [Import and Export](#) ▾ ⚙ [Dashboard Display Settings](#)

[Add filter](#)

[Search](#)

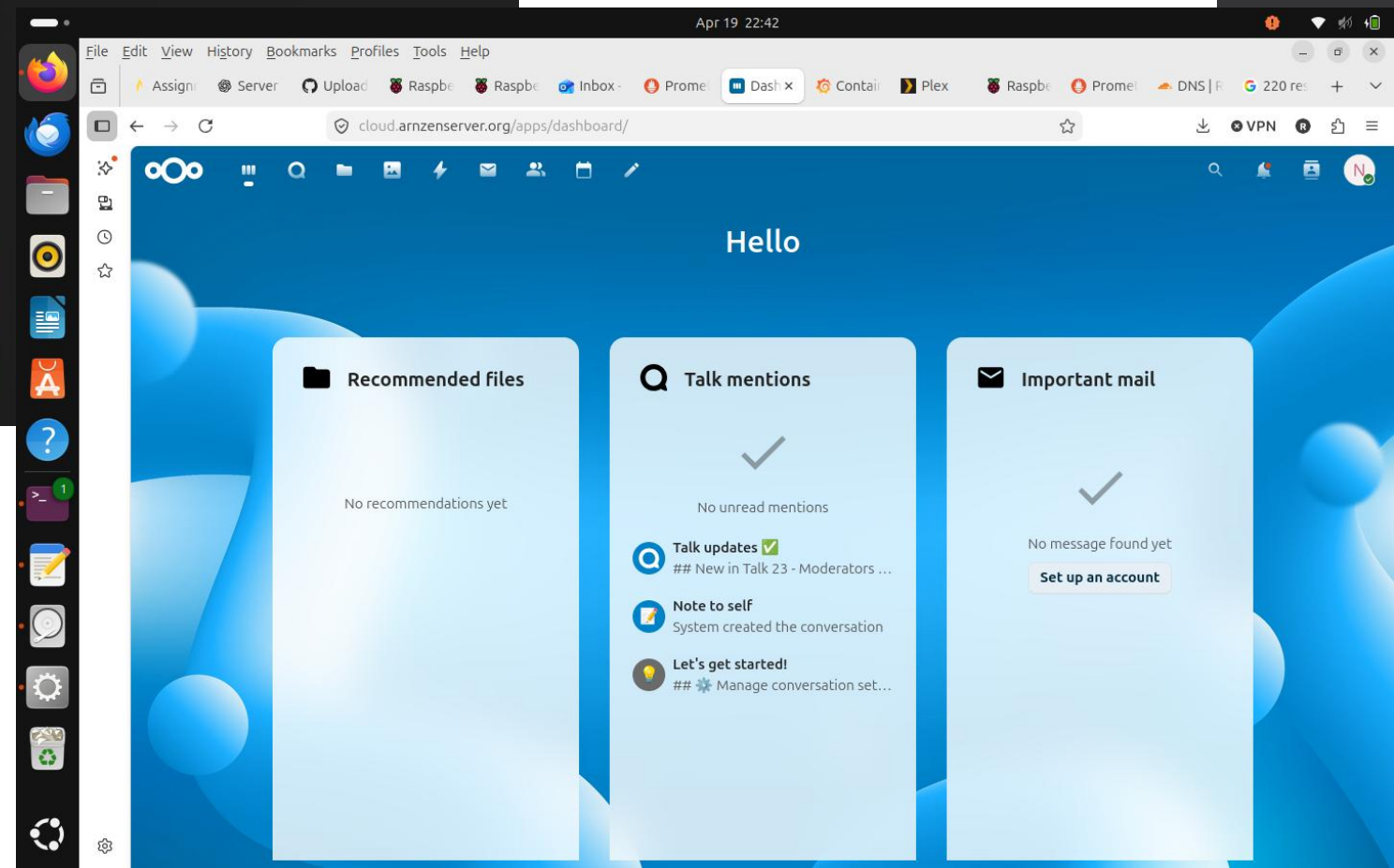
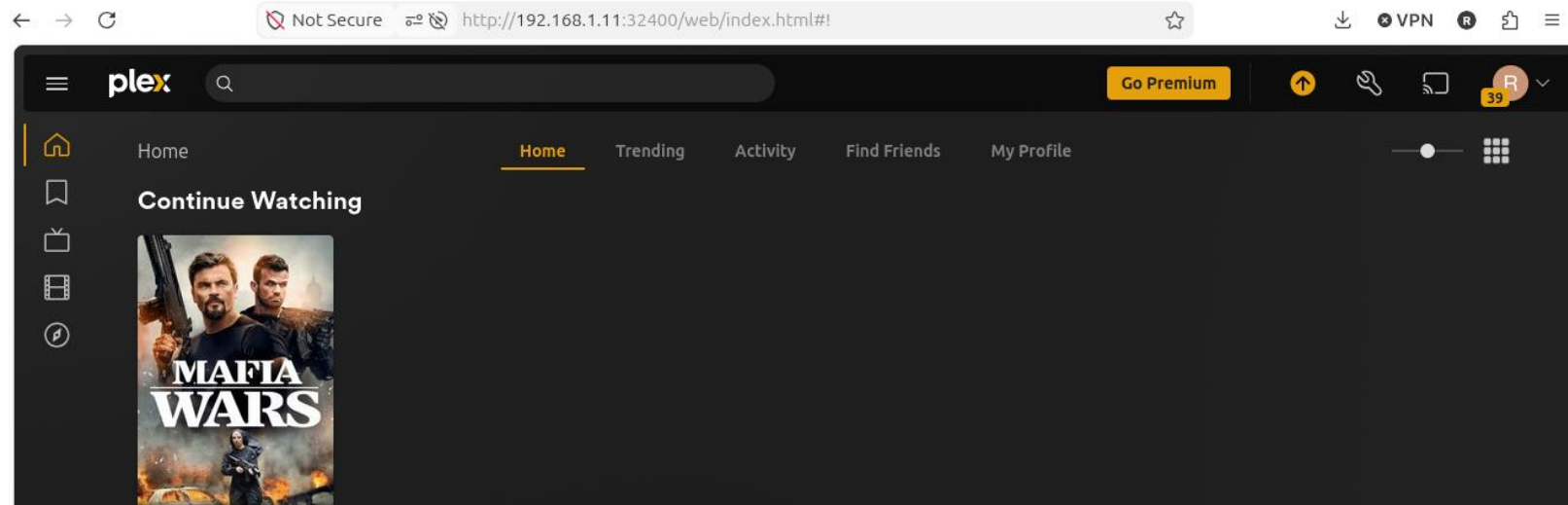
[+ Add record](#)

☐	Type ⓘ	▲	Name ⓘ	Content ⓘ	Proxy status ⓘ	TTL ⓘ	Actions
☐	Tunnel		cloud	homelab	 Proxied	Auto	Edit ▶
☐	Tunnel		media	homelab	 Proxied	Auto	Edit ▶

Cloudflare Nameservers

Every DNS zone on Cloudflare is assigned a set of Cloudflare-branded nameservers.

Type	Value
NS	giancarlo.ns.cloudflare.com
NS	meiling.ns.cloudflare.com



Monitoring

Monitoring services:

- Prometheus for metrics collection
- Grafana for dashboards
- node_exporter for host metrics
- cAdvisor for container metrics

What is monitored:

- CPU usage, RAM usage, disk usage, uptime
- container memory, container CPU, container count

[Select scrape pool](#) [⌵](#)[Filter by target health](#) [⌵](#)[Filter by endpoint or labels](#)

cadvisor

1 / 1 up 🟢 [^](#)

Endpoint

Labels

Last scrape

State

<http://cadvisor:8080/metrics>

instance="cadvisor:8080"

job="cadvisor" [▼](#)[🕒](#) 15.572s ago [▼](#)UP[📄](#) 178ms

node_exporter

1 / 1 up 🟢 [^](#)

Endpoint

Labels

Last scrape

State

<http://node-exporter:9100/metrics>

instance="node-exporter:9100"

job="node_exporter" [▼](#)[🕒](#) 27.355s ago [▼](#)UP[📄](#) 225ms

prometheus

1 / 1 up 🟢 [^](#)

Endpoint

Labels

Last scrape

State

<http://prometheus:9090/metrics>

instance="prometheus:9090"

job="prometheus" [▼](#)[🕒](#) 9.728s ago [▼](#)[📄](#) 10msUP

Grafana Results

Node Exporter dashboard shows:

- disk usage
- root filesystem usage
- uptime
- total root filesystem size

Container dashboard shows:

- total memory usage
- container count
- per-container CPU usage

- per-container memory usage
- Plex CPU and memory
- Nextcloud CPU and memory

- Home
- Bookmarks
- Starred
- Dashboards

Playlists

Snapshots

Library panels

Shared dashboards
- Explore
- Drilldown
- Alerting
- Connections

Add new connection

Data sources
- Administration

☆

Edit

Export v

Share v

<<

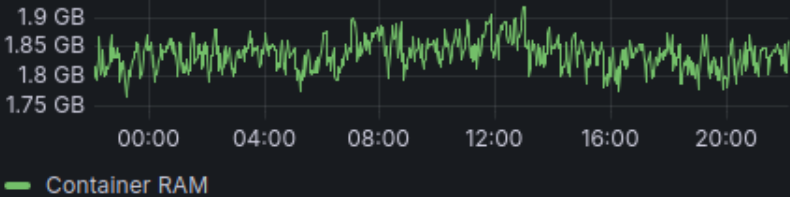
🕒 Last 24 hours v

>>

🔍

🔄 Refresh v

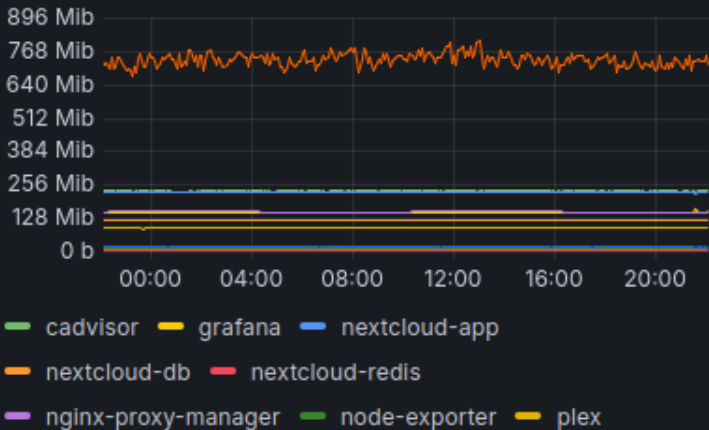
Total Memory Usage



Container Count

10

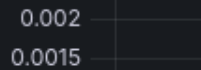
Container Memory



Container CPU



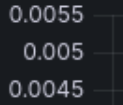
NextCloud CPU



NextCloud RAM



Plex CPU



Plex Memory



Raspberry Pi

Current design:

- Pi polls health/monitoring data
- Python script controls LEDs
- LEDs represent server conditions

LED mapping:

- Green = server reachable
- Blue = Plex service healthy
- Yellow = resource warning
- Red = critical issue / disk or service problem

LED Hardware setup

- Wired outputs:
- GPIO17 → Green
- GPIO27 → Blue
- GPIO19 → Yellow
- GPIO26 → Red

Circuit pattern:

- GPIO → resistor → LED anode
- LED cathode → GND

Learning With AI

AI was used to:

- assist research
- help troubleshooting
- help document and track changes
- to create python code

Workflow: ask AI for guidance → Ask questions → implement manually → test the system → document the verified result

AI mainly helped with Docker/Linux troubleshooting, reverse proxy design, monitoring setup, and Raspberry Pi coding

Challenges and Fixes

- Challenges:
- getting remote access correct
- understanding reverse proxy vs direct service access
- debugging Grafana “No data” panels
- deciding between LCD display and LEDs
- Raspberry Pi reflash / setup issues

Fixes:

- validated Plex remote access directly
- kept Nextcloud on reverse proxy path
- confirmed Prometheus targets and Grafana datasource
- switched to simpler LED hardware strategy
- rebuilt the Pi from a clean flash

Future Improvements

- extend Raspberry Pi from LEDs to LCD/OLED display
- add alerts/notifications
- expand monitoring into predictive maintenance
- potentially connect with the separate ML monitoring project later

Questions?